Linear Convergence of Gradient Descent

An End-to-End Meta-Math Walkthrough

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Abstract

Smoothness turns a gradient step into a quantified decrease. Strong convexity both guarantees an optimum and links the remaining objective gap to the gradient size. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

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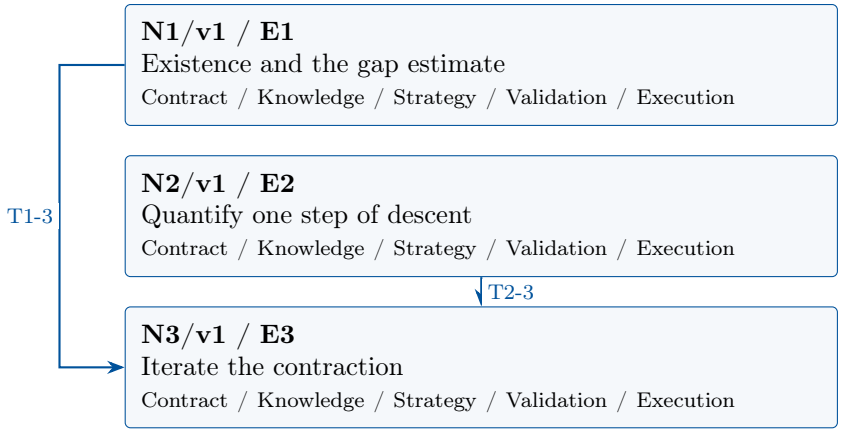
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *Let $f : \mathbb{R}^d \to \mathbb{R}$ be C^1, μ-strongly convex, and have an L-Lipschitz gradient, with $0 < \mu \le L$. Strong convexity here means $f(y) \ge f(x) + \langle \nabla f(x), y - x\rangle + (\mu/2)\|y - x\|^2$ for every x, y. Prove there is a unique minimizer x_* and, for exact-arithmetic iterates $x_{k+1} = x_k - \nabla f(x_k)/L$, $f(x_k) - f(x_*) \le (1 - \mu/L)^k (f(x_0) - f(x_*))$. Also $x_k \to x_*$. Interpret the bound at $k = 0$ as equality, including when $\mu = L$.*

A. Foundations and permitted imports

Use the fundamental theorem of calculus along a segment, Cauchy–Schwarz, and attainment of the minimum of a continuous function on a compact set. The gradient Lipschitz property is $\|\nabla f(x) - \nabla f(y)\| \le L\|x - y\|$. All norms are Euclidean. The theorem concerns exact iterations and does not certify rounding-error behavior.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1-3: N1/E1/v1 to N3. A unique x_* exists, $\nabla f(x_*) = 0$, and $f(x) - f(x_*) \leq \|\nabla f(x)\|^2/(2\mu)$.

Receiving activity. N3.1

Compatibility. E1/v1 and E2/v1 for every iterate. Check: Objective convergence alone is not automatically convergence of points; the quadratic growth estimate supplies that bridge.

T2-3: N2/E2/v1 to N3. $f(x - \nabla f(x)/L) \leq f(x) - \|\nabla f(x)\|^2/(2L)$.

Receiving activity. N3.1

Compatibility. E1/v1 and E2/v1 for every iterate. Check: Objective convergence alone is not automatically convergence of points; the quadratic growth estimate supplies that bridge.

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Existence and the gap estimate

Object and role. Use a quadratic lower bound for coercivity, then minimize the quadratic support model. **Local obligation:** A unique x_* exists, $\nabla f(x_*) = 0$, and $f(x) - f(x_*) \le \|\nabla f(x)\|^2/(2\mu)$.

Inputs. Strong convexity with $\mu > 0$.

C — Contract

Publish E1/v1 only after establishing this obligation: A unique x_* exists, $\nabla f(x_*) = 0$, and $f(x) - f(x_*) \le \|\nabla f(x)\|^2/(2\mu)$.

K — Knowledge

Strong convexity with $\mu > 0$. Retain the stated assumptions and import versions.

S — Strategy

Use a quadratic lower bound for coercivity, then minimize the quadratic support model.

V — Validation

The lower bound must grow at infinity; a merely convex function need not attain its infimum.

E — Execution

Carry out activities N1.1–N1.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N1.1 The strong-convexity inequality at $x = 0$ gives $f(y) \ge f(0) + \langle \nabla f(0), y\rangle + (\mu/2)\|y\|^2 \to \infty$ as $\|y\| \to \infty$. Choose a ball outside which $f > f(0)$; compact attainment inside it gives a global minimizer. Differentiability gives zero gradient there. Strict convexity, which follows from strong convexity, gives uniqueness (or apply the quadratic lower bound at either minimizer).

N1.2 For $g = \nabla f(x)$, strong convexity at x and $y = x_*$ gives $f(x_*) \ge f(x) + \langle g, x_* - x\rangle + (\mu/2)\|x_* - x\|^2 \ge f(x) - \|g\|^2/(2\mu)$ by completing the square. Rearrangement proves the gap estimate.

Export E1/v1. A unique x_* exists, $\nabla f(x_*) = 0$, and $f(x) - f(x_*) \le \|\nabla f(x)\|^2/(2\mu)$.

Acceptance gate. The lower bound must grow at infinity; a merely convex function need not attain its infimum.

Targeted repair. If the sign of the gradient-gap estimate is reversed, complete the square explicitly; the resulting bound is a lower bound on gradient size in terms of the gap. Withhold this export and recheck its descendants after a versioned correction.

B. N2/v1: Quantify one step of descent

Object and role. Integrate the gradient difference along a segment to derive the descent lemma.
Local obligation: $f(x - \nabla f(x)/L) \le f(x) - \|\nabla f(x)\|^2/(2L)$.

Inputs. An L-Lipschitz gradient and $s = -\nabla f(x)/L$.

C — Contract

Publish E2/v1 only after establishing this obligation: $f(x - \nabla f(x)/L) \le f(x) - \|\nabla f(x)\|^2/(2L)$.

K — Knowledge

An L-Lipschitz gradient and $s = -\nabla f(x)/L$. Retain the stated assumptions and import versions.

S — Strategy

Integrate the gradient difference along a segment to derive the descent lemma.

V — Validation

The factor $1/2$ comes from integrating t from zero to one; the step is exactly $1/L$.

E — Execution

Carry out activities N2.1–N2.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N2.1 For any vector s, the fundamental theorem gives $f(x+s) - f(x) - \langle \nabla f(x), s\rangle = \int_0^1 \langle \nabla f(x + ts) - \nabla f(x), s\rangle \, dt$. Its absolute value is at most $\int_0^1 Lt\|s\|^2 dt = (L/2)\|s\|^2$.

N2.2 Substitute $s = -\nabla f(x)/L$. The linear term is $-\|\nabla f(x)\|^2/L$ and the quadratic term is $\|\nabla f(x)\|^2/(2L)$, giving the claimed decrease.

Export E2/v1. $f(x - \nabla f(x)/L) \le f(x) - \|\nabla f(x)\|^2/(2L)$.

Acceptance gate. The factor $1/2$ comes from integrating t from zero to one; the step is exactly $1/L$.

Targeted repair. If arbitrary positive step sizes are declared descending, retain the smoothness quadratic term and verify its coefficient before choosing a step. Withhold this export and recheck its descendants after a versioned correction.

C. N3/v1: Iterate the contraction

Object and role. Combine the gap-gradient inequality with one-step descent, then control distance by strong convexity at the minimizer. **Local obligation:** The stated geometric objective bound holds and $\|x_k - x_*\|^2 \le 2(f(x_k) - f(x_*))/\mu \to 0$.

Inputs. E1/v1 and E2/v1 for every iterate. Incoming proof exports: E1/v1 at N3.1, E2/v1 at N3.1.

C — Contract

Publish E3/v1 only after establishing this obligation: The stated geometric objective bound holds and $\|x_k - x_*\|^2 \le 2(f(x_k) - f(x_*))/\mu \to 0$.

K — Knowledge

E1/v1 and E2/v1 for every iterate. Incoming proof exports: E1/v1 at N3.1, E2/v1 at N3.1. Retain the stated assumptions and import versions.

S — Strategy

Combine the gap-gradient inequality with one-step descent, then control distance by strong convexity at the minimizer.

V — Validation

Objective convergence alone is not automatically convergence of points; the quadratic growth estimate supplies that bridge.

E — Execution

Carry out activities N3.1–N3.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N3.1 Put $\Delta_k = f(x_k) - f(x_*) \ge 0$. E2 and E1 give $\Delta_{k+1} \le \Delta_k - \|\nabla f(x_k)\|^2/(2L) \le (1 - \mu/L)\Delta_k$. Induction gives the geometric bound. If $\mu = L$, the first step has gap zero and all later iterates are the minimizer.

N3.2 Strong convexity at x_*, where the gradient is zero, gives $\Delta_k \ge (\mu/2)\|x_k - x_*\|^2$. The objective bound therefore implies convergence of the iterates themselves. This completes the existence, rate, and convergence claims.

Export E3/v1. The stated geometric objective bound holds and $\|x_k - x_*\|^2 \le 2(f(x_k) - f(x_*))/\mu \to 0$.

Acceptance gate. Objective convergence alone is not automatically convergence of points; the quadratic growth estimate supplies that bridge.

Targeted repair. If point convergence is inferred solely from function values, insert the positive quadratic growth bound with its factor $\mu/2$. Withhold this export and recheck its descendants after a versioned correction.

IV. CONTINUOUS MATHEMATICAL PROOF

A. N1 — Existence and the gap estimate

1. **N1.1.** The strong-convexity inequality at $x = 0$ gives $f(y) \geq f(0) + \langle \nabla f(0), y\rangle + (\mu/2)\|y\|^2 \to \infty$ as $\|y\| \to \infty$. Choose a ball outside which $f > f(0)$; compact attainment inside it gives a global minimizer. Differentiability gives zero gradient there. Strict convexity, which follows from strong convexity, gives uniqueness (or apply the quadratic lower bound at either minimizer).

2. **N1.2.** For $g = \nabla f(x)$, strong convexity at x and $y = x_*$ gives $f(x_*) \geq f(x) + \langle g, x_* - x\rangle + (\mu/2)\|x_* - x\|^2 \geq f(x) - \|g\|^2/(2\mu)$ by completing the square. Rearrangement proves the gap estimate.

B. N2 — Quantify one step of descent

1. **N2.1.** For any vector s, the fundamental theorem gives $f(x+s) - f(x) - \langle \nabla f(x), s\rangle = \int_0^1 \langle \nabla f(x+ts) - \nabla f(x), s\rangle\, dt$. Its absolute value is at most $\int_0^1 Lt\|s\|^2 dt = (L/2)\|s\|^2$.

2. **N2.2.** Substitute $s = -\nabla f(x)/L$. The linear term is $-\|\nabla f(x)\|^2/L$ and the quadratic term is $\|\nabla f(x)\|^2/(2L)$, giving the claimed decrease.

C. N3 — Iterate the contraction

1. **N3.1.** Put $\Delta_k = f(x_k) - f(x_*) \geq 0$. E2 and E1 give $\Delta_{k+1} \leq \Delta_k - \|\nabla f(x_k)\|^2/(2L) \leq (1 - \mu/L)\Delta_k$. Induction gives the geometric bound. If $\mu = L$, the first step has gap zero and all later iterates are the minimizer.

2. **N3.2.** Strong convexity at x_*, where the gradient is zero, gives $\Delta_k \geq (\mu/2)\|x_k - x_*\|^2$. The objective bound therefore implies convergence of the iterates themselves. This completes the existence, rate, and convergence claims.

□

V. WORKED ORCHESTRATION

For $f(x, y) = (x^2 + 4y^2)/2$, take $\mu = 1$, $L = 4$. The iteration sends (x, y) to $(3x/4, 0)$. From $(1, 1)$ the first objective drops from $5/2$ to $9/32$, below the certified upper bound $(3/4)(5/2) = 15/8$. Later the actual remaining gap shrinks by $9/16$ per step, so the theorem's $3/4$ factor is conservative.

A. Boundary and hypothesis audit

An arbitrary large step fails even for $f(x) = x^2/2$. With step size 3 the iteration is $x_{k+1} = -2x_k$, and the objective grows by a factor of four unless $x_0 = 0$. The prescribed step and smoothness bound cannot be discarded.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Rejected argument, diagnosis, and correction

Rejected N3 argument: decreasing objective values guarantee convergence to the unique optimum with a geometric rate. Monotone numbers alone provide neither conclusion. **Diagnosis:** Strategy omitted the quantitative gradient-gap bridge. **Repair:** use E1 to obtain the actual contraction factor and the strong-convexity distance estimate. This completes the original theorem under its existing assumptions; recheck both objective and point convergence.

B. Publication and reuse

The corrected export retains its explicit assumptions. A hypothetical revision receives a new version, and each consuming activity is rechecked; this scenario is not a recorded runtime event. **Next direction:** Investigate accelerated methods or inexact gradients with newly stated error bounds; do not transfer this exact-arithmetic rate unchanged.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading. Boyd and Vandenberghe, Convex Optimization for convexity, optimality, and descent; University of Maryland, Scientific Computing I for the numerical-analysis setting. The estimates used here are derived explicitly in §4.